Localising Copernicus Air-Quality Forecasts for Arnavutköy, Istanbul: Next-Day Forecasts and Health Alerts

Updated Project Proposal (Concept Note) — Assignment 1, revised after instructor feedback (October 2026)
MTH4800 Innovation Campus AI Applications · Capstone: AI for the SDGs · Group 9
GitHub repository: github.com/maliksonmez/arnavutkoy-air-quality-forecast (original submission kept in *docs/*)

What changed after feedback

- **External data added:** hourly meteorology (wind speed/direction, temperature, humidity, precipitation, pressure) and Copernicus CAMS air-quality forecasts (PM10, PM2.5, NO2, SO2, Saharan dust) for the station location [8–10].
- **New framing (as suggested):** instead of forecasting from scratch, we **correct the daily CAMS forecast** with local station measurements and weather. CAMS underestimates NO2 at the station by about two-thirds but follows its day-to-day changes (r = 0.63).
- **Localisation:** wind-direction analysis (airport sector) and a Saharan-dust event analysis.
- **Result:** the corrected model reduces validation MAE by 34% (PM10), 12% (PM2.5) and 45% (NO2) against "tomorrow = today", and by 40–58% against raw CAMS; evaluated days rose from 48–49 to 129–147 of 164.
- **Focus:** research questions reduced to three; time zone of the station data verified; leakage rules stated explicitly.
- **Planned for A2 (not yet done):** neighbouring SİM stations, İBB hourly traffic density for NO2 and Sentinel-5P NO2. Large construction sites were not analysed separately, because their locations and activity periods are not available to us.

1. Project Name

Localising Copernicus Air-Quality Forecasts for Arnavutköy, Istanbul: Next-Day Forecasts and Health Alerts

2. Overview

Air pollution is a major urban health risk, especially for children, older adults and people with asthma or heart disease. Monitoring portals only report past measurements, and the free Copernicus (CAMS) forecast (~11 km) cannot see local sources such as rush-hour traffic. We combine both: a global forecast of tomorrow's large-scale weather and transport, corrected by what the local station measures.

> *"Our tool helps residents of Arnavutköy — especially sensitive groups such as children, older adults and people with asthma or heart disease — plan their next day by correcting the Copernicus air-quality forecast with local measurements and warning them when WHO guideline levels are likely to be exceeded."*

- **SDG link: SDG 11.6** (urban air quality; indicator 11.6.2: PM2.5/PM10) is the primary target, **SDG 3.9** (illness and death from air pollution; indicator 3.9.1) is secondary [7].
- **Target users:** residents of Arnavutköy and sensitive groups; local decision-makers as secondary users.
- **How we will know it works:** lower MAE than *both* "tomorrow = today" and the raw CAMS forecast on a held-out test period, and the share of WHO-exceedance days that the tool correctly flags.

3. Background

The WHO 2021 guidelines recommend 24-hour mean levels of **15 µg/m³ for PM2.5, 45 for PM10, 25 for NO2 and 40 for SO2** [1]. At this station the daily mean exceeded them on 36% of valid days for PM2.5 and NO2 and 13% for PM10 (SO2: none). Istanbul studies show that traffic is a dominant urban source [3], that black carbon follows NO2 with rush-hour peaks [4] and that meteorology strongly affects particle levels [5]. Recent ML work starts from simple baselines and uses lagged pollutant values as key inputs [2]. Correcting a physical model with local observations (statistical post-processing) is a standard way to localise such forecasts [8].

4. Key Objectives

4.1 Research Questions

1. Can a model that corrects the CAMS forecast with local measurements and weather forecast next-day concentrations more accurately than both persistence and raw CAMS?
2. How do wind direction and local sources (the airport to the north-east, traffic and the urban area to the south-east) shape pollution at the station?
3. How much do Saharan-dust events contribute to PM10 and to WHO exceedances, and can CAMS help anticipate them?

4.2 Key Steps and User Flow

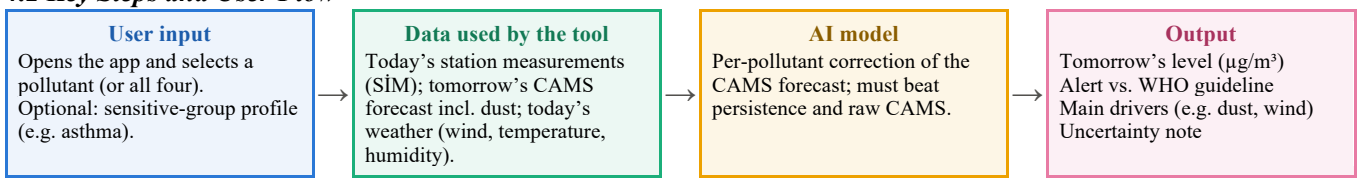


Figure 1. What the user provides and what the tool produces. The development pipeline is documented in the repository (*docs/flowchart.png*).

Key steps. A1: problem, data, EDA, external data, first correction experiment (completed). A2: Ridge vs. Random Forest/XGBoost, walk-forward validation, simple end-to-end Streamlit tool. A3: user-profile testing, ethics, deployment, final application.

5. Methods and Workflow

5.1 Datasets

- **Station (target):** SİM İstanbul–Arnavutköy (operated by İBB), 41.2209° N, 28.7075° E; 16.09.2023–16.09.2026, 26,305 hourly records of PM10, PM2.5, SO2 and NO2 (µg/m³) [6].
- **CAMS forecasts:** CAMS European air-quality forecast (~11 km, hourly; ensemble of European models), retrieved for the station point via the Open-Meteo Air Quality API [8, 9]; 100% coverage of the study period.
- **Meteorology:** hourly ERA5-based weather from the Open-Meteo Historical Weather API [9, 10]: wind speed and direction (converted to u/v components), temperature, humidity, precipitation, pressure.

5.2 Data Cleaning, Preprocessing and Exploratory Analysis

- **Station data (as in A1):** 3.9–5.8% missing, long gaps not filled; seven "985" readings set to missing; valid day = ≥ 18 hourly values; daily **median** as target, daily **mean** for WHO comparison.
- **Alignment:** SİM timestamps are offset by 56 seconds (e.g. 00:00:56) and were floored to the hour before merging. **Time zone verified:** the station–CAMS correlation peaks at a 0-hour shift for NO2 and PM10, so both are on local time.
- **CAMS vs. station:** CAMS captures day-to-day variation (daily r = 0.70 PM10, 0.55 PM2.5, 0.63 NO2) but is biased low: on average 61% of the measured PM10 and only 32% of the measured NO2. It also misses the morning rush-hour NO2 peak (Fig. 2a) — exactly the local signal the station adds.
- **Wind direction:** wind blows from N–NE (the airport sector, ~7 km) in 50% of hours, and these are the **cleanest** hours (median NO2 14.8 µg/m³). Winds from ESE–S — the direction of the adjacent multi-lane road and the urban area — are rarer (11%) but bring about three times more NO2 (46.7) and twice the PM10 (Fig. 2b); the pattern holds in every season and wind-speed class. An airport signal is therefore not visible at this station.
- **Saharan dust:** 20 days (10 episodes, mostly March–May) have a CAMS daily dust mean above 20 µg/m³. On these days the median PM10 is 64 vs. 23 µg/m³, the PM2.5/PM10 ratio drops from 0.61 to 0.44 (coarse mineral particles) and 65% exceed the WHO PM10 guideline (other days: 11.5%) (Fig. 2c).

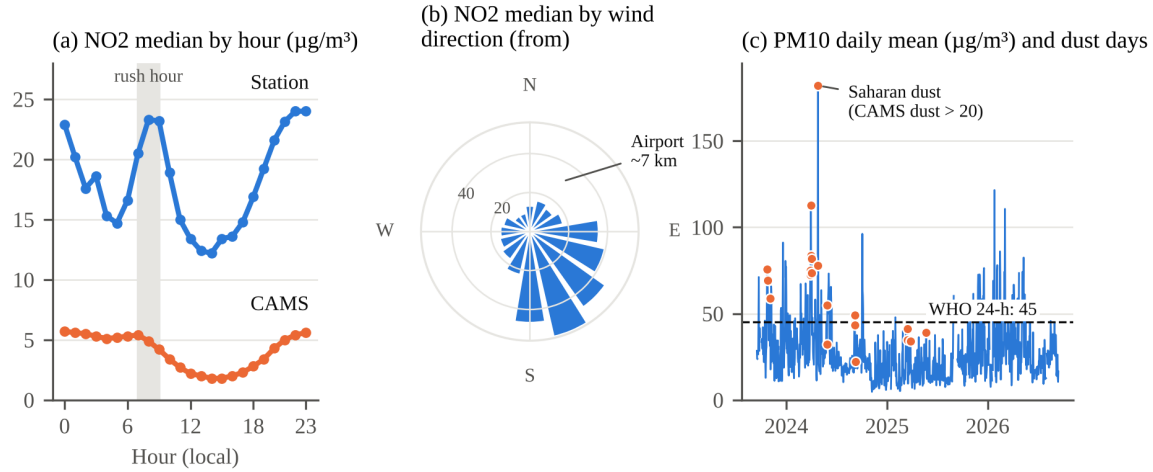


Figure 2. (a) Hourly medians of NO2: station vs. CAMS; CAMS is lower and misses the 07–09 peak. (b) Median NO2 by wind direction (bars point to where the wind comes from); the airport sector is the cleanest. (c) PM10 daily mean (≥ 18 hourly values) against the WHO guideline; orange points mark days with CAMS dust above 20 µg/m³.

5.3 Modelling and Experiments

Task and split (unchanged, for comparability): forecast the next day's daily median per pollutant; chronological split — training to 06.07.2025, validation 07.07–17.12.2025, test from 18.12.2025 (**still unused**, kept for the final evaluation). The A1 results were first reproduced exactly. **Model:** one Ridge regression per pollutant (standard scaling; alpha chosen by 5-fold time-series cross-validation inside the training period only, no longer on the validation set). **Inputs:** the pollutant's own median today, yesterday and seven days before the target; today's weather and share of hours with ESE–S wind; CAMS daily mean (and dust for PM); weekday and season of the target day. With each pollutant's own history, SO2 gaps no longer remove days

for the others. **Leakage rules:** no input from the target day except CAMS, which is itself a forecast. **E1** uses only information available at the end of today (today's CAMS). **E2** (CAMS post-processing) adds CAMS for the target day.

Model (validation MAE, µg/m³; lower is better)	PM10	PM2.5	SO2	NO2
Persistence ("tomorrow = today")	7.62	2.23	**1.39**	9.89
Raw CAMS forecast	8.72	3.26	5.20	13.14
CAMS with linear bias correction only	8.14	7.00	5.32	7.88
E1: Ridge, station + weather + today's CAMS	7.12	2.44	1.58	8.68
E2: Ridge, station + weather + CAMS for tomorrow	**5.04**	**1.96**	1.65	**5.48**
Days evaluated (of 164), identical for all rows above	144	147	111	129
A1 reference: Ridge with 28 lags vs. persistence (48–49 days)	7.67 / 7.57	3.40 / 2.63	1.51 / 1.19	11.42 / 9.58

Table 1. Validation results, 07.07–17.12.2025. All rows above the line are evaluated on the same days.

Findings. E2 beats persistence and raw CAMS for PM10, PM2.5 and NO2, and is the best model in four to five of the six validation months; persistence wins only in the low-pollution late summer, while the largest gains come in autumn and winter. E1, which uses only information available today, also beats persistence for PM10 (−7%) and NO2 (−12%), so the gain does not rely only on the CAMS lead-time advantage (Section 6). The CAMS-only bias correction fails for PM2.5 because the station's PM2.5 level dropped in the validation period (mean 8.9 vs. 16.4 µg/m³ in training) while CAMS did not; models that also use the latest station value adapt. For SO2, which is low and never exceeds the WHO guideline, no model beats persistence, so SO2 will have lower priority. The strongest inputs are CAMS for tomorrow and today's station value — the global forecast and the local measurement complement each other.

5.4 Deliverables

A **Streamlit web app** with tomorrow's corrected forecast, a WHO-based alert (from the daily mean), dust and wind context, historical trends and SHAP explanations; the **GitHub repository** (notebooks 01–05, processed data, README with a change log); Docker/cloud deployment; final report, presentation and demo video.

6. Limitations and Ethical Considerations

- **CAMS lead time and weather:** the archive keeps the most recent forecast for each hour, so the CAMS value for "tomorrow" in E2 has a shorter lead time than one available today; E2 is therefore optimistic and E1 is the conservative bound. In A2 we will re-evaluate with genuine day-ahead CAMS runs from the Atmosphere Data Store [8]. ERA5 is observed (reanalysis) weather, so only today's weather is used.
- **Single station:** a pilot, not representative of all of Istanbul; neighbouring stations (e.g. Başakşehir, Sultangazi) will be added. Planned: İBB hourly traffic density for NO2, Sentinel-5P NO2 (GIS week).
- **Data quality:** "985" and PM2.5 > PM10 values remain unconfirmed without device information; the PM2.5 level shift needs checking with the operator.
- **Ethics:** the main risk is a wrong "clean air" forecast for a sensitive user. We will show uncertainty, use cautious wording near thresholds, issue no forecast when inputs are missing, state that the tool does not replace official warnings and give no medical advice.

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